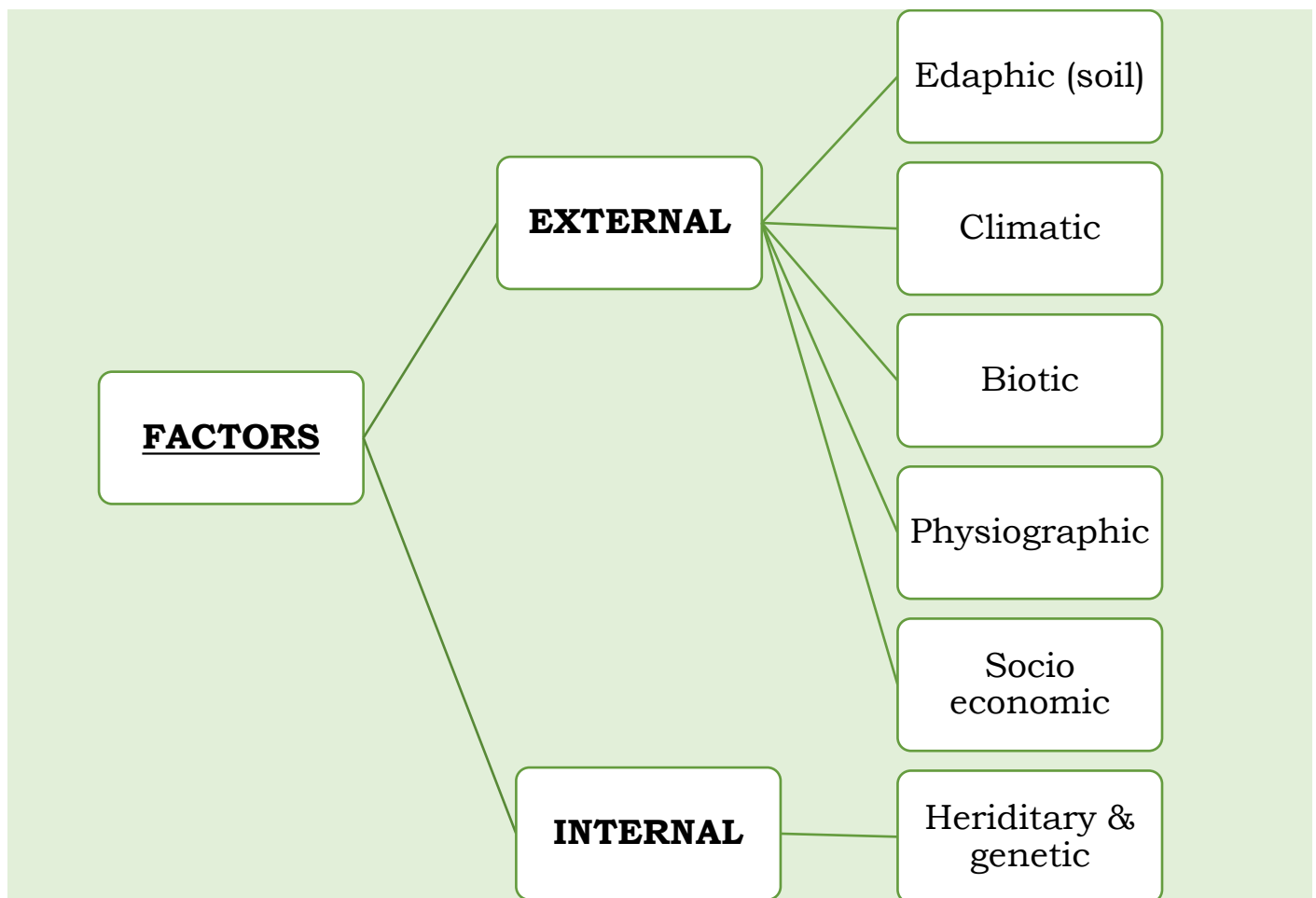


FACTORS AFFECTING CROP PRODUCTION

Crop production is **vital for agricultural economy of our country** which is affected by various factors which are discussed as following



INTERNAL FACTORS:

Internal factors are those factors which talks **about the genetic makeup of the plants/crops. These characters are hereditary in nature** or sometimes comes due to the mutation etc. Internal factors decide the desirable and undesirable characters of a crop plants.

Genetic factors

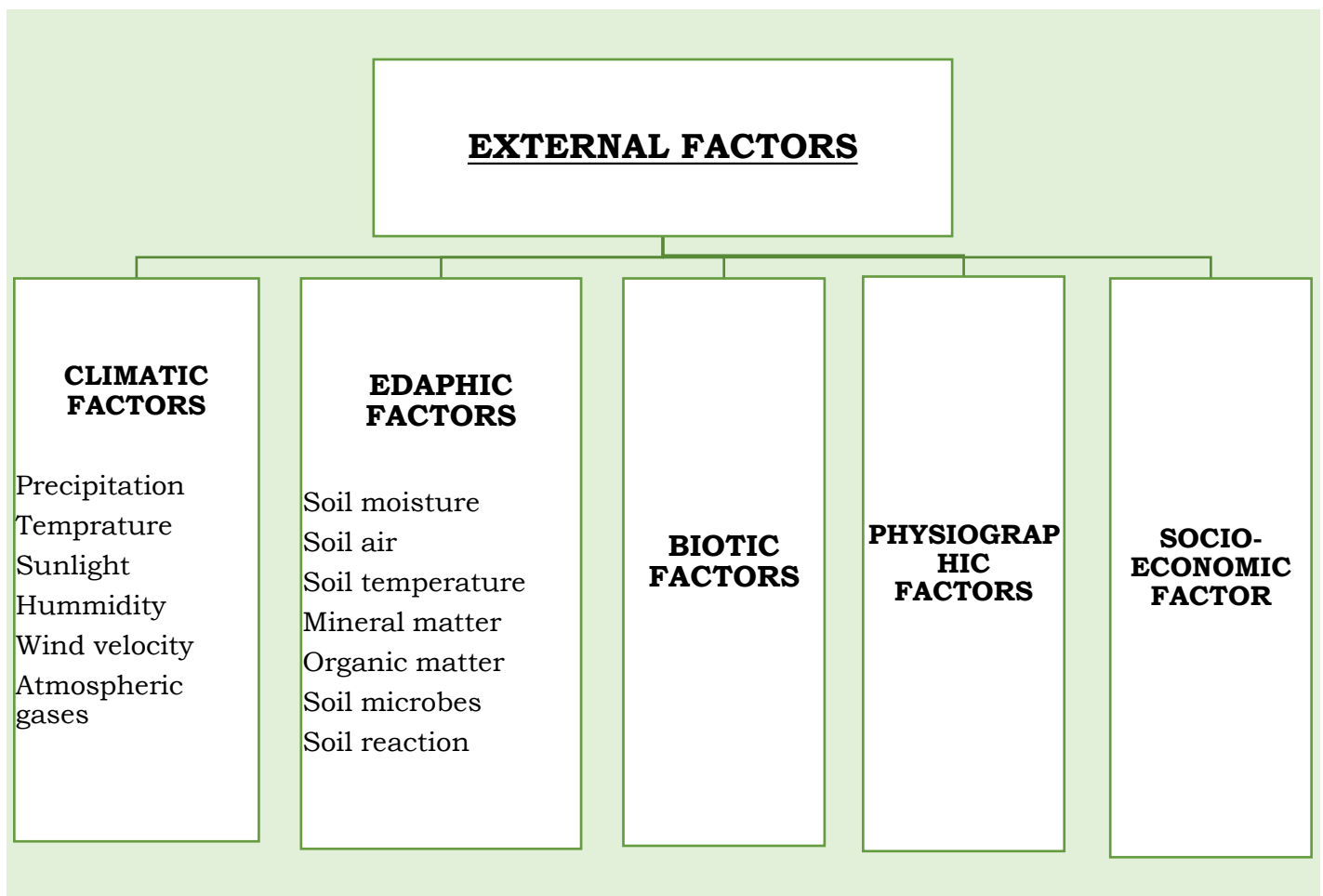
- ✓ High yielding ability
- ✓ Early maturity

- ✓ Resistance to lodging
- ✓ Drought flood and salinity tolerance
- ✓ Tolerance to insect pests and diseases
- ✓ Chemical composition of grains (oil content, protein content)
- ✓ Quality of grains (fineness, coarseness)
- ✓ Quality of straw (sweetness, juiciness)

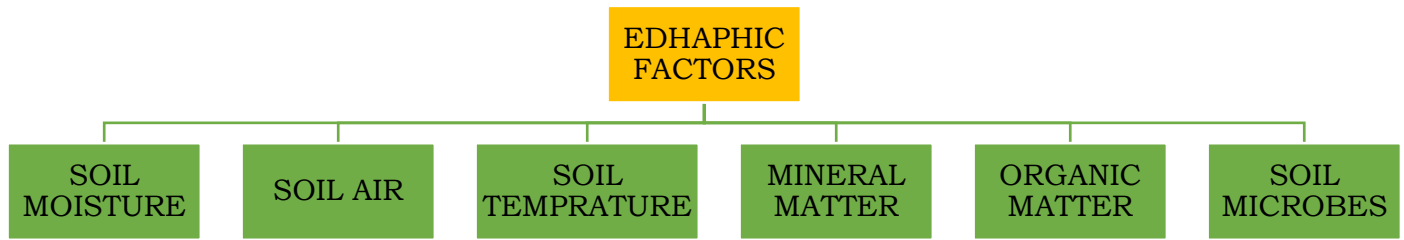
The above characters are less influenced by environmental factors since they are governed by genetic make-up of crop

EXTERNAL FACTORS

These are the factors which impact **the growth & development of the crop plant like root & shoot development, leaf formation, flower formation, yield etc.**



EDAPHIC FACTORS (SOIL) Plants grown **in land completely depend on soil** in which they grow.



SOIL MOISTURE

- ✓ Water is a **principal constituent** of growing plant which it extracts from soil.
- ✓ Water is essential for photosynthesis.
- ✓ The moisture ***range between field capacity and permanent wilting point is available to plants.***
- ✓ Soil water helps in ***chemical and biological activities*** of soil including mineralization.
- ✓ It influences the soil environment E.g. it moderates the soil temperature from extremes.
- ✓ ***Nutrient availability and mobility increase*** with increase in soil moisture content.

SOIL AIR

- ✓ Aeration of soil is absolutely ***essential for the absorption of water*** by roots
- ✓ ***Germination is inhibited*** in the absence of oxygen
- ✓ O₂ is ***required for respiration*** of roots and microorganisms.
- ✓ Soil air is ***essential for nutrient availability*** of the soil by breaking down insoluble mineral to soluble salts.
- ✓ For ***proper decomposition of organic matter.***
- ✓ **Potato, tobacco, cotton linseed, tea and legumes** need higher O₂ in soil air
- ✓ Rice **requires low level of O₂ and can tolerate water logged** (absence of O₂) condition.

SOIL TEMPERATURE

- ✓ It affects the ***physical and chemical*** processes going on in the soil.
- ✓ It influences the ***rate of absorption of water and solutes*** (nutrients)
- ✓ It ***affects the germination of seeds*** and growth rate of underground portions of the crops like tapioca, sweet potato.

- ✓ Soil temperature **controls the microbial activity** and processes involved in the nutrient availability
- ✓ Cold soils are not conducive for rapid growth of most of agricultural crop

SOIL MINERAL MATTER

- ✓ The mineral content of soil is derived from the weathering of rocks and minerals as particles of different sizes.
- ✓ These are the sources of plant nutrients eg; Ca, Mg, S, Mn, Fe, K etc

SOIL ORGANIC MATTER

- ✓ It supplies all **the major, minor and micro** nutrients to crops
- ✓ It **improves the texture** of the soil
- ✓ It **increases the water holding capacity** of the soil,
- ✓ It is a **source of food** for most microorganisms
- ✓ Organic acids released during decomposition of organic matter **enables mineralisation** process thus releasing unavailable plant nutrients

SOIL ORGANISMS

- ✓ The raw organic matter in the soil is decomposed by different microorganisms which in turn **releases the plant nutrients**
- ✓ **Atmospheric nitrogen is fixed by microbes** in the soil and is available to crop plants through symbiotic (Rhizobium) or non-symbiotic (Azospirillum) association

SOIL REACTION (PH)

- ✓ Soil reaction is the pH (**hydrogen ion concentration**) of the soil.
- ✓ Soil pH affects crop growth and neutral soils with pH 7.0 are best for growth of most of the crops
- ✓ Soils may be acidic (7.0)
- ✓ Soils **with low pH is injurious to plants due high toxicity of Fe and Al.**
- ✓ Low pH also **interferes with availability of other plant nutrients.**

BIOTIC FACTORS

Beneficial and harmful effects caused **by other biological organism** (plants and animals) on the crop plants

Plants

- ✓ **Competitive and complimentary** nature among field crops when grown together. When different **crops of cereals and legumes** are grown together, mutual benefit results in higher yield (synergistic effect,).
- ✓ **Competition** between plants occurs when there is demand for nutrients, moisture and sunlight particularly when they are in short supply or when plants are closely spaced.
- ✓ **Competition between weed and crop** plants as parasites eg: Striga parasite weed on sugarcane crop.

Animals

- ✓ Soil fauna like **protozoa, nematode, snails, and insects** help in organic matter decomposition, while using organic matter for their living.
- ✓ **Insects and nematodes cause damage to crop yield** and considered as harmful organisms.
- ✓ **Honey bees and wasps help in cross pollination** and increases yield and considered as beneficial organisms.
- ✓ **Burrowing earthworm facilitates** aeration and drainage of the soil as ingestion of organic and mineral matter by earthworm results in constant mixing of these materials in the soils.
- ✓ **Large animals cause damage to crop plants** by grazing (cattle, goats etc).

PHYSIOGRAPHIC FACTORS

- ✓ **Topography** is the nature of surface earth (levelled or sloppy) is known as topography. Topographic factors affect the crop growth indirectly.
- ✓ **Altitude** – increase in altitude cause a decrease in temperature and increase in precipitation and wind velocity (hills and plains)
- ✓ **Steepness of slope**- it results in run off of rain water and loss of nutrient rich top soil
- ✓ **Exposure to light and wind**- a mountain slope exposed to low intensity of light and strong dry winds may results in poor crop yields (coastal areas and interior pockets).

SOCIO-ECONOMIC FACTORS

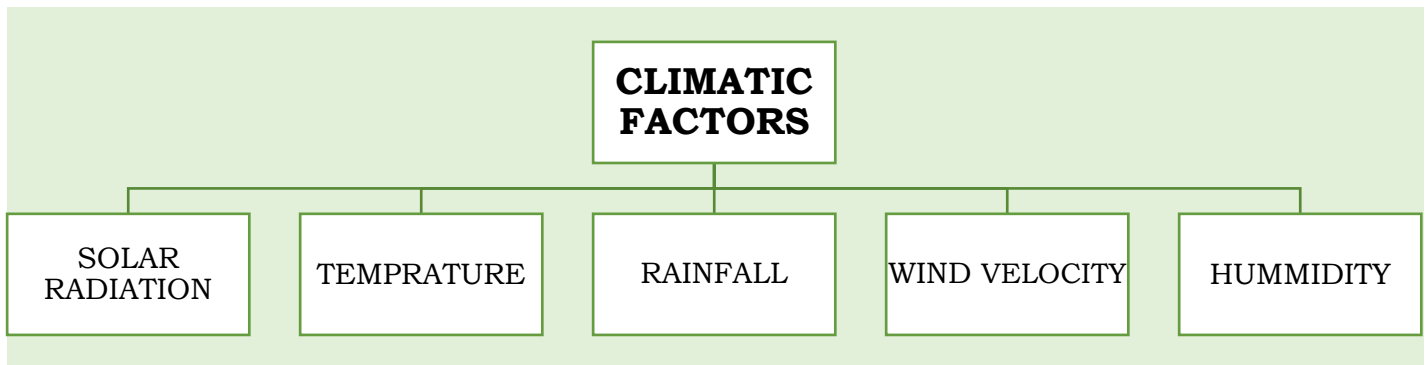
- ✓ **Society inclination** to farming and members available for cultivation.

- ✓ Appropriate **choice of crops** by human beings to satisfy the food and fodder requirement of farm household.
- ✓ **Breeding varieties** by human invention for increased yield or pest & disease resistance.
- ✓ The **economic condition** of the farmers greatly decides the input/ resource mobilizing ability (marginal, small, medium and large farmers).

CLIMATIC FACTORS

Impact Of Climatic Factors On Crop

- ✓ Climate is the **most dominating factor** that impact the suitability of the crops to a particular region. ***The yield potential mainly depends upon the climate.***
- ✓ More than 50% variation in crops yield is due to climatic differences.
- ✓ The most important climatic factor that influence growth & development of crops are - **solar radiation, temperature & rainfall(precipitation).**
- ✓ **Humidity, wind velocity & atmospheric gases** have a very little influence on crop growth.



SOLAR RADIATION

Solar energy is required for **seed germination, leaf expansion, growth of stem & shoot, flowering, fruiting, & physiological** functions of crop plant.

There are 4 types of impact of solar radiation on crops: -

1. **Photosynthetic effect**
2. **Photoperiodic effect**
3. **Thermal effect**

4. Other effect

PHOTOSYNTHETIC EFFECT

Most influential on crop plants as it **impacts the whole process of photosynthesis of crop plants**. The solar radiation influences the production of enzymes useful for photosynthesis, development of photosynthetic apparatus, growth, yield formation & the final yield.

- a. **Enzyme production**-Photosynthesis is **governed by the enzymes PEP (phosphoenol pyruvate carboxylase) & ribulose bisphosphate carboxylase**. Increase in light intensity increase the activity of these enzymes.
- b. **Development of photosynthetic apparatus**- this includes chlorophyll, leaf area, & their exposure.
- ✓ **Chlorophyll- formation** of chlorophyll is promoted in **light of 0.300 to 0.338 μm** .
 - ✓ **Leaf characteristics**- high light intensity decrease the leaf surface area & increase leaf thickness which reduces the rate of photosynthesis. These leaves have high number of stomata **which increases the CO_2 uptake**.
 - ✓ **Light impact leaf orientation**.

c. **Growth**-

Light interception increases with increase in the leaf area index which is essential for growth of plant.

d. **Yield formation**: -

Light impact yield attributes ex low light intensity in groundnut during the flowering reduces the number of flowers per plant hence less yield.

PHOTOPERIODIC EFFECT:

Photoperiodic effect is defined **as the duration of sunlight available** to the plants. The duration of the **sunlight impact various process in crop plant like leaf formation, flowering & floral initiation**.

Based on photoperiodic impact plants are classified into three categories.

1. **Short day plants**- require **less than 10 hours** of sunlight for flower initiation daily ex- rice, sorghum, maize etc.
2. **Long day plant**- these plants require **more than 10 hours** of sunlight for flower initiation daily ex- wheat, barley, oats etc.
3. **Day neutral plants**: - These plants **do not get affected** by the duration of sunlight available for the initiation of flower ex-cotton, sunflower, buckwheat etc.

IMPACTS OF PHOTOPERIOD

- ✓ More or less duration of light in short day and long day plants impact the reproductive stage of plant which **delays or inhibit the flowering**.
- ✓ Photoperiodic effect is **directly interrelated to the temperature during** the particular day ex rice is a short-day plant and **receives less than 10 hr.** of sunlight but does not initiate flowering if subjected to the low temperature.

THERMAL IMPACT –

Received sunlight is converted into heat by the plant. Which impact the **rate of transpiration, increase temperature of leaves & other parts.**

OTHER IMPACT

- ✓ **Assimilation of nutrients**-light intensity impact the assimilation of **the nutrient ex phosphorus accumulation is lowest during darkness & There is increase in activity of nitrate reductase during dark phase. Uptake of potassium is highest** in barley during high intensity.
- ✓ **Translocation of photosynthates**- the light intensity impacts the **translocation** of nutrients from source to sink.

TEMPRATURE

Cardinal Temperature -the minimum, maximum & optimum temperature of a crop are known as a cardinal temperature.

Most of the crops grows best in temp range of **15 °C to 30 °C** & many of crop plant will die at a temperature of **above 45 to 55 °C.**

<u>CARDINAL</u> <u>TEMPERATURE</u> ↓	<u>COOL SEASON CROP</u>	<u>WARM SEASON CROP</u>
Minimum	0 to 5° C	15 to 20° C
Maximum	30 to 38° C	45 to 50° C
Optimum	25 to 30° C	30 to 38° C
Examples →	Wheat, barley, oats etc.	Rice, maize, sorghum etc.

IMPACTS OF TEMPRATURE

1. Biochemical Reaction-

Temp impacts the rate of biochemical **reaction as with every 10° C rise in temperature the rate of biochemical** reaction gets doubled.

2. Uptake Of Carbon Dioxide-

Optimum temperature for CO₂ uptake is 24° C for wheat & barley. As the temperature increases above the optimum temperature CO₂ uptake decreases due to increase in stomatal & mesophyll resistance.

3. Rate of photosynthesis-

Low & high temperature exposure of crop plant even for short duration reduces the rate of photosynthesis.

4. Development of photosynthetic apparatus-

Low & high temp causes degradation of chlorophyll hence yellowing of leaves. Temperature also governs the rate of leaf emergence & leaf expansion.

5. Influence on growth hormones-

Low & high temperature then the optimum temperature causes the disbalance in the growth hormone i.e. auxin, gibberellin, cytokinin etc. which shows the inhibitory effect on the growth of plants.

6. Dry matter production-

Temperature more or less than the optimum temperature reduces the accumulation of the dry matter in the crop plants hence reduce yield.

7. Impact germination, leaf initiation, tillering, Flowering, spikelet initiation & grain filling.

8. Yield and yield formation-

Impact grain formation, grain filling, grain yield as high temperature inhibits the starch synthesis in the crops.

9. Pest & disease-

Influence the growth & multiplication of **insect's ex aphid their growth starts with high temperature with humid climate and reduction in population** with low & dry temperature. Early blight disease of potato is caused by low temperature stress.

HIGH TEMPERATURE STRESS

- ✓ **Mineral nutrition-** reduce absorption & assimilation of nutrients ex absorption of calcium is reduced **in maize at temp above 28° C.**
- ✓ **High temperature-** high temperature impact the root elongation, shoot elongation in wheat & rice, reduce plant height.
- ✓ **Pollen development-** High temperature during booting stage causes pollen abortion.

LOW TEMPERATURE STRESS-

Mainly occur due to frost at high altitude.

- ✓ **Survival-** Frost or very low temperature during early stages of growth of plants reduces the chances of survival.
- ✓ **Cell division & cell elongation-**causes retardation **in cell division & elongation** hence reduced growth of the plant.
- ✓ **Photosynthesis-** at low temperature **the activity of pyruvate diaphosphatase** is reduced hence low rate of photosynthesis.

- ✓ **Water transport**- low temperature **reduce the permeability of the cells due to freezing which restricts the entry of water into plant cell**. This will impact various physiological process of the crop plants.
- ✓ **Growth**- Temperate **crops prefer low temperature for vegetative growth & tropical** crop growth is retarded in low temperature.

RAINFALL

Rainfall is one of the most **important weather factors that influence the crop growth directly or indirectly**. **Rainfall is** the source of soil moisture under rainfed condition hence have huge impact on rainfed crops.

Impacts

1. Nutrient management-

Response to applied **nutrients depends upon soil moisture during high rainfall leaching of the nutrients occur or during dry spell mobility** of the nutrients reduces an optimum amount of moisture is required in soil for availability of nutrients. Hence rainfall impact the nutrient uptake by plants.

2. Weed management –

Rainfall impacts the weed seed reserve in the soil which causes the germination of weed seed after rainfall **ex *Chenopodium album*** after heavy rainfall in June.

3. Emergence of pest-

Heavy rainfall followed by the break in monsoon with heavy sunshine result in multiplication of pest which causes infestation in crops.

4. Disease-rainfall

Favors the growth of **various disease-causing microorganism** in crops ex ergot infection in sorghum, Alternaria leaf blight on sunflower.

5. Crop lodging-

Heavy rainfall followed by the wind of **speed more than 25Km/hr.** causes crop lodging which negatively impact the grain yield, spikelet formation & harvesting operation.

6. Produce quality-

Quality of the produce reduces **if rainfall occur at the time of ripening or harvesting** ex immature grains, reduced size of grain like blackened tip of barley, black fungal disease of wheat grain etc.

RELATIVE HUMMIDITY

Relative humidity directly influences **the water relations & indirectly influences the leaf growth, photosynthesis, pollination, occurrence of disease and yield.**

1. Water relation-

Directly impact the evapotranspiration rate hence impact the water requirement of crops.

Ex dry region -plant faces high water stress due to low RH.

2. Photosynthesis-

Low RH increase the evapotranspiration which causes water deficit in crops. This water **deficit causes partial & full closure of stomata** hence causes the reduction in rate of transpiration.

3. Pollination-

Pollen viability increases with the higher RH while low RH increases the desiccation of pollen grains

4. Pest & disease-

High incidence of pest & disease under high humidity condition. High RH favors the germination of spores of microorganism at a very fast rate. **Ex the severity of powdery mildew on grapes increases with increase in humidity of 85%.**

✓ **Scab disease of wheat** increase with high RH.

✓ **Sheath rot incidence** in rice is also dependent on the RH.

5. Grain yield-

Very high or very low RH is not conducive for high grain yield ex high RH negatively impact the grain yield in maize. Wheat grain yield is also reduced with high RH.

WIND VELOCITY

1. **Hot wind increases the PET** in plants which causes desiccation of crop by replacing the humid air with dry air.
2. **Increases cuticular transpiration** then stomatal transpiration due to increased wind velocity.
3. **Disrupt the balance of growth hormone**- wind increases the ethylene production in barley & rice or reduces the gibberellic acid contents of root & shoot of rice.
4. High wind during the month of June **July causes tearing of rice leaves**.
5. High wind **causes lodging of crops** hence reduce yield.